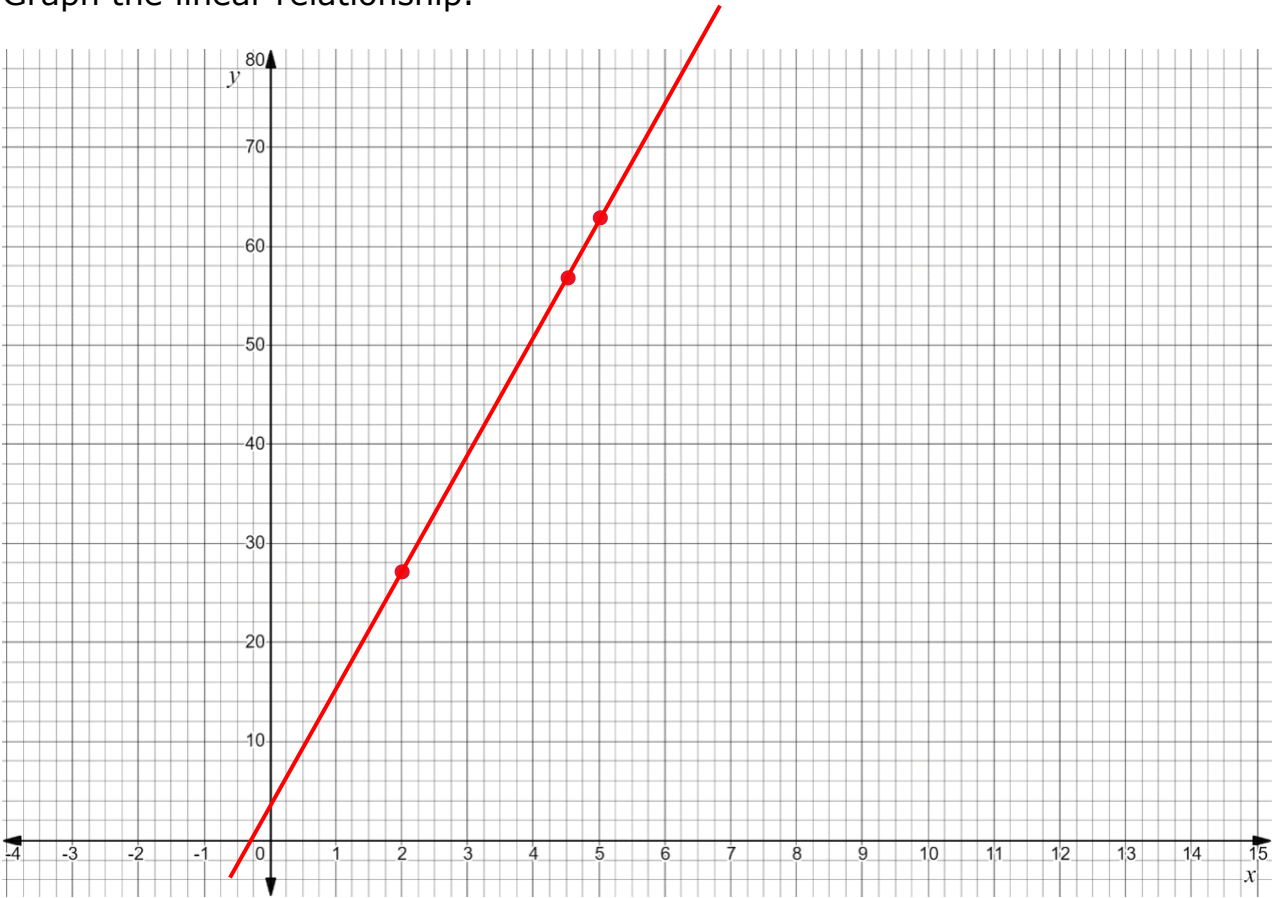


The table of values shows the linear relationship between x , the height of water in a container, and y , the weight of the container.

Height of Water (inches)	2	4.5	5	7.25	8
Weight of Container (ounces)	27	57	63	90	99

1. Graph the linear relationship.



2. Place a check in the box of a key feature that can be determined from the table of values.

Domain	Range	x -intercept	y -intercept	Slope	Increasing/ Decreasing
✓	✓			✓	✓

3. The equation $12x - y = -3$ models the linear relationship. Place a check in the box of a key feature that can be determined using the equation.

Domain	Range	x -intercept	y -intercept	Slope	Increasing/ Decreasing
✓	✓	✓	✓	✓	✓

4. Write the equation $12x - y = -3$ in slope-intercept form.

$$-y = -12x - 3$$

$$y = 12x + 3$$

5. What domain makes sense for the context?

$0 \leq x < \infty$ or until the max capacity for the container is reached.

6. What range makes sense for the context?

$3 \leq x < \infty$ or until the max weight for the container is reached.

7. Determine the x -intercept.

$$0 = 12x + 3$$

$$-3 = 12x$$

$$x = -\frac{1}{4}$$

8. Interpret the x -intercept.

When the height of the water is $-\frac{1}{4}$ inches, the container will weigh 0 oz. (which is impossible- meaning the bucket will always weigh 3 oz., even without water in it).

9. Determine the slope.

$$y = mx + b$$

$$m = 12$$

10. Interpret the slope.

Every inch of water added to the container increases its weight by 12 oz.